Energy Research & Social Science 121 (2025) 103959

Contents lists available at ScienceDirect

Energy Research & Social Science

journal homepage: www.elsevier.com/locate/erss

Perspective

Models like heroes? Making Integrated Assessment Models (IAMs) ready for deep decarbonization and a socio-economic transformation

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ARTICLE INFO

Keywords:
Integrated assessment modeling
Mitigation
Climate change
Justice
Critical modeling
Socio-economic transformation
Politics of knowledge

ABSTRACT

Integrated Assessment Models (IAMs) are among the most influential tools for global climate action. While IAM research contributed enormously to the current awareness of the climate crisis, their approach to transformation pathways has faced significant criticism. This critique illuminates that diverse aspects of social, political, and economic processes shape IAM research. This paper provides the first systematic literature review (based on the PRISMA approach) of the diverse criticisms that specifically target the political implications of IAM modeling practices. After reviewing a sample of 71 scientific publications extracted from the Web of Science and Scopus databases, we describe the various social science concepts utilized in the literature and argue that they reveal stabilizing functions performed by IAM modeling. These functions legitimize and rationalize the crisis-ridden societal conditions, if they are not actively addressed in the research process. To elucidate the modus operandi of these stabilizing functions, we introduce the concept of ideology as a mediator between existing societal conditions and the research practice, offering a novel framework to conceptualize the critique of IAM modeling practices. To illustrate how IAMs often fall short of their goal of presenting pathways out of the climate crisis, instead contributing to the stabilization of crisis-ridden societal conditions, and how to prevent this, we introduce an analogy to superheroes. Based on these findings, we suggest a participatory approach that includes the visions of social movements and systematic research on deep socio-economic transformations facilitated by the Categorical Utopia framework.

1. Introduction

In the midst of a worsening climate crisis, and despite years of efforts by scientists to explore options for action for decision-makers, limiting global warming in accordance with the Paris Climate Agreement remains out of reach. In this perspective article, we assume that this deep decarbonization is only possible through a profound socio-ecological transformation (Deep Transformation).

With respect to climate mitigation, Integrated Assessment Models (IAMs) are the most influential scientific tool, as they form the basis of the Working Group on Mitigation (WG3) of the Intergovernmental Panel on Climate Change (IPCC). IAMs vary in their philosophy, architecture, and are typically tailored to a specific question [1], and are thus used in several contexts. Their commonality lies in the integration of social and physical phenomena. For example, by modeling the energy system, agriculture, and the economy. Furthermore, in their ambition to "help understand how possible actions or choices might lead to different future outcomes" ([2], p. 51). This makes IAMs important and influential tools to inform policy makers about possible mitigation pathways, but also about the future in general. This includes questions arising from the potential failure of mitigation. By addressing these questions, the IPCC, and the IAM community at the forefront, have made a major contribution to the recognition and understanding of climate change by policymakers and society at large.

However, when it comes to exploring transformation pathways, IAM research has often been criticized by the social sciences in general and by Science and Technology Studies (STS) in particular. At the forefront, the focus on technological solutions, normative assumptions, sociological aspects of science, and the lack of transparency of these are criticized. This article, first reviews this critique on IAMs that aims at the political aspects of IAM research based on a systematic literature review following the PRISMA flow [3]. Based on these findings, we argue that this leads to a practical crisis of IAM research. We will develop a metaphor herein: IAMs are compared to Hollywood superheroes, fixating on defeating one villain, like emissions, while overlooking the bigger picture. Despite their heroic efforts, IAMs and superheroes overlook the broader systemic issues, and thus they mark the prevailing crisis-ridden societal conditions as rational and legitimate. In other words, IAMs examine futures to find solutions to mitigate the climate

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https://doi.org/10.1016/j.erss.2025.103959
Received 12 September 2024; Received in revised form 27 January 2025; Accepted 28 January 2025
Available online 11 February 2025
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crisis, but the way they do it stabilizes the status quo and thus the path into the climate crisis. Subsequently, we discuss the role of ideology in this context and conceptualize the exercised critique. Finally, possible ways forward and the first steps towards systematic research into Deep Transformations are outlined.

2. Methods

2.1. Positionality

All the authors involved identify as cis-male, white and have so far experienced limited effects of the climate crisis and benefit economically from the global division of labor and the power relations between the Global North and South. Felix Krawczyk is convinced that the climate crisis requires a profound social transformation that encompasses all areas of life. He is a climate activist and therefore has insights into the various demands and attitudes within the climate justice movement. He is trained as a physicist and climate scientist but also has a background in environmental humanities such as political ecology.

Andreas Braun is accredited as a geoecologist, sociologist and has acquired some limited further training in philosophy, economy, and law science, hence, he is an interdisciplinary scientist. Although not an activist himself, Andreas acknowledges and supports the important role played by environmental and climate activist engagements, as long as they adhere to non-violence and do not transgress the boundaries of peaceful civil disobedience. He has taken a clear supportive stance of such movements in media communication.

2.2. Definitions

We understand critique as a normative endeavor, as critique must always apply a set of values. For example, the measure can aim at accuracy, explanatory power, instrumental usefulness, or emancipatory potential.

The critique can be descriptive, reflexive, and constructive [4]. In the tradition of Critical Theory, we relate critique that aims at an isolated deficit to the overall societal conditions. By the term societal conditions, we denote the dominant logic of a society that structurally suggests certain types of behavior and forms of relationships towards human and non-human-nature. Therefore, the notion includes, for example, the economic structure, cultural norms, political settings, and discourses. We perceive this dominant logic as a logic of exclusion, i.e. it is structurally suggested to satisfy needs at the expense of other humans [5] or non-human-nature. In this sense, we call futures that maintain the current societal conditions projections. In contrast, we understand utopias as a possible reconfiguration of the existing and the potential [6] towards other societal conditions.

From the perspective of Critical Theory, ideology is a necessary product of social conditions to justify existing injustices [7]. Critique of ideology is not only about criticizing the actual social issues but also their perception [7]. It is, therefore, an epistemic and normative problem at the same time. Furthermore, this means that the critique of ideology is not only aimed at the ideology (epistemic) but also at the practice with which it is reproduced (emancipatory change) [7]. In this sense, our critique is intended to stimulate a fundamental reformation of IAM modeling and research practice.

2.3. Method

The critique of IAMs was identified through a systematic literature review according to a PRISMA scheme adapted by us [3]. We thereby sought out literature whose criticism focused on the political aspect of IAMs or their use in science. A strict distinction between this and technical criticism is not possible, as IAMs are socio-technical hybrids [8]. Therefore, technical criticism with implications for research into possible futures was also included. Furthermore, the review focuses on the critique of IAMs and less on the critique of policies such as the implementation of Negative Emissions Technologies (NETs) being researched with IAMs. We included critique on NETs as far as they are related to IAM research. However, critique that primarily focuses on policies or political implications of NETs is not part of this review. Peer-reviewed science communication articles and modeling studies were also excluded. A total of 71 articles, published between 2001 and 2023, were analyzed. The literature search is limited to the scientific databases Scopus and Web of Science. This pre-selection limits our study to knowledge classified as scientific. Furthermore, this study is limited to analyses that explicitly identify themselves as critiques of IAMs, as dictated by our search query. This is an important limitation of this study. A detailed description of the method and the PRISMA Flow Diagram can be found in the Supplementary Information.

3. Literature review

3.1. IAMs and the background to their scientific genesis

Most IAMs originated in the 1990s but have their roots in the World3 model of the Club of Rome and other climate-economic models that emerged after the oil crisis [9,10]. Before scientists developed the first IAMs for the climate context, models were used for policy advice regarding acid rain. According to van Beek et al. [10], these models have significantly shaped "agenda-setting, target-setting, and evaluation of abatement strategies" (p. 3). When climate change became a pressing issue in the 1980s, the first teams began developing IAMs to support political decisions. Two of these models (ASF, IMAGE) were then used to develop emission scenarios (SR90) for the first IPCC report (FAR). Since the Kyoto Conference in 1997, the significance within the IPCC and the policy relevance of IAMs have continued to rise [10]. Since the Copenhagen Accord (COP15, 2009), they have evolved into the foundation of Working Group 3 (Mitigation) [10]. This is why IAMs are criticized in reference to predecessor models, UN climate conferences, and the IPCC [9].

Over time, an epistemic community has formed around integrated assessment modeling, consisting of modelers, scientists, and decision-makers. This community is not only formed around specific knowledge but around certain practices and intertwined relationships [1,11]. According to Cointe et al. [1], the IAM community forms around the common goal of policy relevance. The community is also characterized by shared norms, infrastructure, space (spatial and intellectual), procedures, and resources that are well adapted or have adapted to the broader research and funding context, keeping it together. The community is organized into institutions such as the Integrated Modeling Consortium (IAMC), Modeling Intercomparison Projects (MIPs), and scenario databases [1]. All these institutions aim to create policy relevance across different model approaches and philosophies [1]. The personnel diversity within the IAM community is criticized, as the development of IAMs largely occurs in the Global North (e.g., [12]). Additionally, women, Black, Indigenous, and People of Color are underrepresented [13,14]. The aforementioned epistemic community is shaped by a specific epistemic culture [11]. This is defined by "affinity, necessity, and historical coincidence" and determines "how we know what we know" ([11], p. 1).

In this context, Ellenbeck and Lilliestam [14] highlight that policymakers are more likely to fund teams that share a common way of discussing issues, known as a "discursive formation". This involves using similar language, concepts, and ideas when approaching a problem. Such alignment is often sought after, as it allows the quantitative results of models to be leveraged in legitimizing political strategies or policies [14]. The technologies represented in models are preferred by policymakers because there is quantitative data about them, and they are formalized in a certain way [13].

At the same time, the structure of IAMs generally favors technological solutions over social change and, in turn, certain technologies over others [15,16]. For example, all scenarios of the Special Report 1.5 Degrees (IPCC SR1.5) are based on NETs [13]. Even though, NETs are heavily criticized for being a moral hazard as they are based on economic considerations rather than feasibility and justice [13,17–21].

3.2. Co-production of knowledge and society

Many articles argue that the production of society and knowledge occurs in a reciprocal relationship [9,18,22–25]. Various frameworks have attempted to understand these interdependencies. We refer to these diverse but implicationally similar frameworks simplistically as co-production here. According to Jasanoff [26], how we understand the world and how we represent it cannot be separated from how we want to live. In sustainability science, this is evident, as there always needs to be a vision of how things could be better. Scientific knowledge is embedded in social practices, identities, norms, conventions, discourses, instruments, and institutions, which, in turn, shape and embed them [26]. In this context, nature is not denied its ontological existence, but it is explained that human access to it depends on a specific world with interpretive authorities and scientific practices that, in turn, shape them. Here, Demeritt [25] notes that early global models are embedded in a specific framing. For example:

- Within this framing, climate change is caused by universal physical laws (GHG) and not by political structure or moral failure.
- These objective facts can be determined by experts.
- The best method to understand the problem is through mathematical models.

This makes it clear that the assumptions of the IAM community, contrary to Pindyck [27] claims, are not arbitrary but are in a reciprocal relationship with societal discourses [14] and can be inherited within the scientific community [28]. This involves not only explicitly value-based assumptions but also technical assumptions that reflect values and are not adequately discussed [18,29,30]. Crucial here is that the results of IAMs depend heavily on the initial conditions [31], and therefore, this context of origin must be reflected. A co-productionist approach can make taken-for-granted assumptions visible by examining the political and normative connections between science and political power [9].

Researchers like Beck and Mahony [24] apply such a co-production lens to IAMs and the IPCC, using the example of NETs. They conclude that assessments are descriptions of the world and instructions on how to rationally manage it. The framing of the problems causes and effects of the climate crisis, and the examination of solutions are mutually influential. This phenomenon becomes problematic when science is tasked by politics to scientifically examine problems, and the problem formulation is uncritically adopted ("Politics of Anticipation") [24]. This is embedded in a scientific concept in which science produces neutral knowledge in the first step [32]. In a separate second step, this knowledge is used by policymakers. Thus, science acts as separated from the social and power relations [26]. This obscures the fact that implicit political and ethical assumptions are made during model development and use.

IAMs contribute to this context through boundary work [24,33,34]. When categories or elements are taken for granted by scientists, they slide into the domain of nature and are thus legitimized as unchangeable and "not available for reconfiguration by new knowledge" ([26], p. 31). In these processes of naturalization, one can always ask, "Who is empowered through knowledge, and to what ends?" ([26], p. 33). Some formations, such as relations of class and capital, and distribution of wealth (e.g., see Table 1: Negishi Weights), are presented in IAMs and assessments in general as unchangeable. The structure of funding also favors scientific research, contributing to a kind of "environmental determinism" that makes societal and, especially, economic conditions seem natural and unchangeable ([25], p. 318).

Beck and Krueger [35] note that IAMs are not just mediators of power but also have agency in the sense of Actor-Network-Theory. In this context, pathways and scenarios have a "world-making" power, potentially shaping the world in their own image and creating new political realities ([24], p. 1). In this light, it is not surprising that McLaren and Markusson [23] observe the co-evolution of technological promises, modeling, policies, and climate change targets. In other words, framings of climate targets in international politics have developed alongside models and science-based technological promises. This has opened up ways for politics, interpreting them as necessary socio-ecological transformations into the expansion of new technologies ("policies of prevarication") ([23], p. 392). These are mediated by the respective modeling practices. Thus, a causal loop is formed, as the models shape the goals, necessary policies, and promised technologies. Numerous studies show this connection with the development and use of Shared Socioeconomic Pathways (SSPs) [1,14,23,36–41].

Among other things, through the SSPs, it is possible to explicitly examine overshoots — scenarios in which the temperature or forcing target is exceeded but then decreases by the end of the century [23]. In the construction of scenarios and models, recourse to narratives is inevitable [14,42]. These narratives serve an epistemic function by connecting the model with the world it represents [42]. Thus, models and scenarios can tell different stories depending on the assumptions made and the elements highlighted. However, these assumptions are not random but embedded in worldviews, raising the question, "[w]hose stories about climate change are systematically excluded from IAM narratives – and thus policy?" ([42], p. 12). These seemingly neutral scientific processes have a significant impact on society and the knowledge made available to it. However, there are epistemic ambiguities in the IAM community and beyond. Scientists sometimes view IAMs as real depictions or independent worlds [43], while their significance is only achieved through human interpretation — referred to as representation [44]. This dual meaning arises from the dual function of IAMs as producers of visions and suppliers of robust recommendations for action [43].

4. Discussion

4.1. Map-makers vs. Superheros

> "Although modellers repeatedly emphasize their scenarios are merely illustrative and their practice should be conceived as 'map-making' [...] they still 'determine what is on the map' and thereby which futures are conceivable or not" ([16], p. 5).

The "map-maker-navigator" metaphor describes the (perceived) division of labor between research (map-maker) and policymaking (navigator). While the map-maker-navigator metaphor has been central to understanding IAMs, it falls short in capturing their diversity and broader socio-political impacts. It inadequately addresses their societal roles and ideological dimensions, and it perpetuates a false dichotomy between science and policy, as IAMs often blur these boundaries rather than maintain them [43]. The superhero metaphor helps address these limitations by illustrating how IAMs, like superheroes, possess distinct strengths and weaknesses [54], operate within specific political and economic contexts, and tend to stabilize the status quo rather than driving systemic change [55]. This metaphor also introduces a normative perspective, emphasizing that superheroes conserve the status quo through "easy solutions" enabled by their superpowers, whereas other approaches may aim to transform societal conditions. Similarly, IAMs and superheroes fulfill a stabilizing function: they 'save the world' by preserving the status quo. In doing so, they reinforce existing power structures, such as property regimes and the interests of the powerful [55]. Furthermore, superheroes confront extreme situations with unrealistic means [55], much like IAMs through the proposed

Table 1
Overview of economic assumptions.

Function	Short Description
Representative agent	"Predominantly utilitarian, albeit unrealistic narrative of human behavior, grounded in assumptions of an idealized lifestyle based on consumerism, accumulation, and high-energy use" ([13], p. 4). Conceptions of being, worldview, and the social are reduced to the Western Homo Economicus [13,45,46].
Damage functions	Damage functions provide monetary losses depending on the global mean temperature. Losses are assigned a monetary value, excluding other forms of valuation [47–49]. Irreversible damages can be financially compensated [47].
Negishi weights	Under the assumption that geographic redistribution is impractical or even impossible [13], the weighting freezes the geographic income distribution [29,50]. It benefits countries that are already richer but have contributed the most to historical emissions.
Discounting rates	Assuming economic growth and society's preference for short-term consumption, the same consumption is less valuable in the future [13,29,48]. This is reflected in positive discounting rates. Reduction of intergenerational justice is shifted to future generations in terms of consumption and costs [13].
Perfect foresight	Assumption that investments are made when economically most sensible and this is known in the future [48]. This leads to the perceived result that a stable carbon pricing policy is the most efficient mitigation option.
GDP growth	Economic growth as normative goal in all states. It does not matter how rich and "developed" they already are. The feasibility of absolute decoupling of growth and emissions is doubted [51]. Growth, therefore, means higher energy consumption and models simulate more negative emissions to compensate for this [52]. Asefi-Najafabady et al. [48] criticize "that IAMs play a key role in distracting attention from the feasibility of societies and economies built around the assumption of limitless economic growth" (p. 8).
Equilibrium theories	These assumptions are controversial in economic science. Heterodox economics advocate instead for "a focus [...] on dynamic change, strong uncertainty, power relations, and social and economic structure beyond aggregated individual agency" ([47], p. 665). Equilibrium theories inadequately describe the complexity of social phenomena and can lead to overly optimistic, risky, or erroneous results [21].
Demand side change	Although reducing energy consumption is an obvious lever, its implementation and effects are questionable [21,53]. For example, lower energy demand pathways result in a lower share of renewable energy, instead of reduced carbon removal [53].

use of NETs. By contrast, the map-maker-navigator metaphor assumes all maps are neutral and of equal value, which oversimplifies the complex and often contested roles of IAMs. Superheroes and IAMs are not intentionally designed to be the way they are but are instead the result of existing power relations, which they also reproduce through ideological elements. In this way, the superhero metaphor also creates a connection to the production of ideology in the cultural realm and therefore allows future studies to draw on the numerous studies of critical media theory aiming at the criticism of ideology.

4.2. Models like heroes?

Therefore, in the following section, we examine the superhero metaphor in detail and delve deeper into its ideological aspects. The characteristics of the superheroes are illustrated in Fig. 1. The reviewed literature demonstrates that the knowledge production with IAMs is shaped by social, economic, and political processes. These processes are shaped by the here and now and are therefore also embedded in existing power relations, habits, and dominant norms. These processes are not sufficiently reflected and included in the research, so elements are inscribed in the research that hinder Deep Transformation. The various social-scientific concepts employed in the literature to criticize IAMs highlight the different functions of those processes. Therefore, we call them *stabilizing functions*. These are summarized in Table 2 and their interactions are illustrated in Fig. 2. IAMs, like superheroes, target a problem (villain) in which they were developed. For other aspects, they are more or less blind. Therefore, the remaining elements of the status quo are rationalized and legitimized. By not incorporating the context of knowledge creation, the superhero aspect of IAMs remains unaddressed.

Both also follow the same problem-solving sequence. Once the villain is defeated, the world is back in order. The superhero or the solution only comes into play when the crisis is acute to prevent negative outcomes. Therefore, they operate under the mode of the politics of the negative [56]. In other words, superheroes become active in the moment when society perceives a situation as having no viable alternatives other than risk politics. When the crisis is over, they can leave again, until the next mission is needed. Therefore, it is not about a longer-term transformation, but rather about fixing an error.

The context around the superhero is ignored in two ways. First, the context is not part of the problem. Even though projections are based on different scenarios, root causes – in terms of what dynamics and societal conditions have led to the rise of the villain – are not questioned. Therefore, a change in context is also not part of the solution. Second, the context, or the societal conditions, are supposed to remain unchanged, because except for the influence of the villain, everything should remain as it is. This dynamic is reflected in several aspects of IAM research. Firstly, IAMs are still based on the same instrumental human-nature relationship. In the past, it was thought that we could simply emit CO_2 into the atmosphere, but today it is believed that we can take it out again. In addition, there is the misconception that nature can be technically calculated and a sense of entitlement, i.e. the attitude of being able to configure the earth system as desired. Finally, the same power relations are taken for granted and are rearticulated in the production of knowledge.

This is an example of the functioning of ideology, which Roy Bhaskar refers to as the "There is no Alternative Formation" ([57], p. 38). Such a formation arises when a mistaken belief is initially presented as necessary. However, this notion of inevitability is inaccurate, as there are indeed other options available [58,59]. If this notion is accepted, it involves "defensive or supporting mechanisms to sustain the original false necessity and build a defensive shield" ([60], p. 114). Jaeggi [7, p. 281] describes ideology in this sense as normative modes of world perception. They normatively define the options for possible actions and thus fundamentally determine what is to be done [7, p. 281]. The reviewed literature does not engage with the concept of ideology. Even though the examined articles highlight that knowledge and societal visions are produced collaboratively, ideology explains why precisely these visions exist and not others. Moreover, it embeds the existing criticism in the societal context. This means that the phenomena criticized in the examined articles can only be fully explained by underlying societal conditions that produce and require ideology. This results in a closed-loop process as illustrated in Fig. 2. Ideology influences the entire research process by legitimizing and justifying the existing societal conditions by artificially shrinking the space of possibilities. This shrinking is characterized by the assumption that

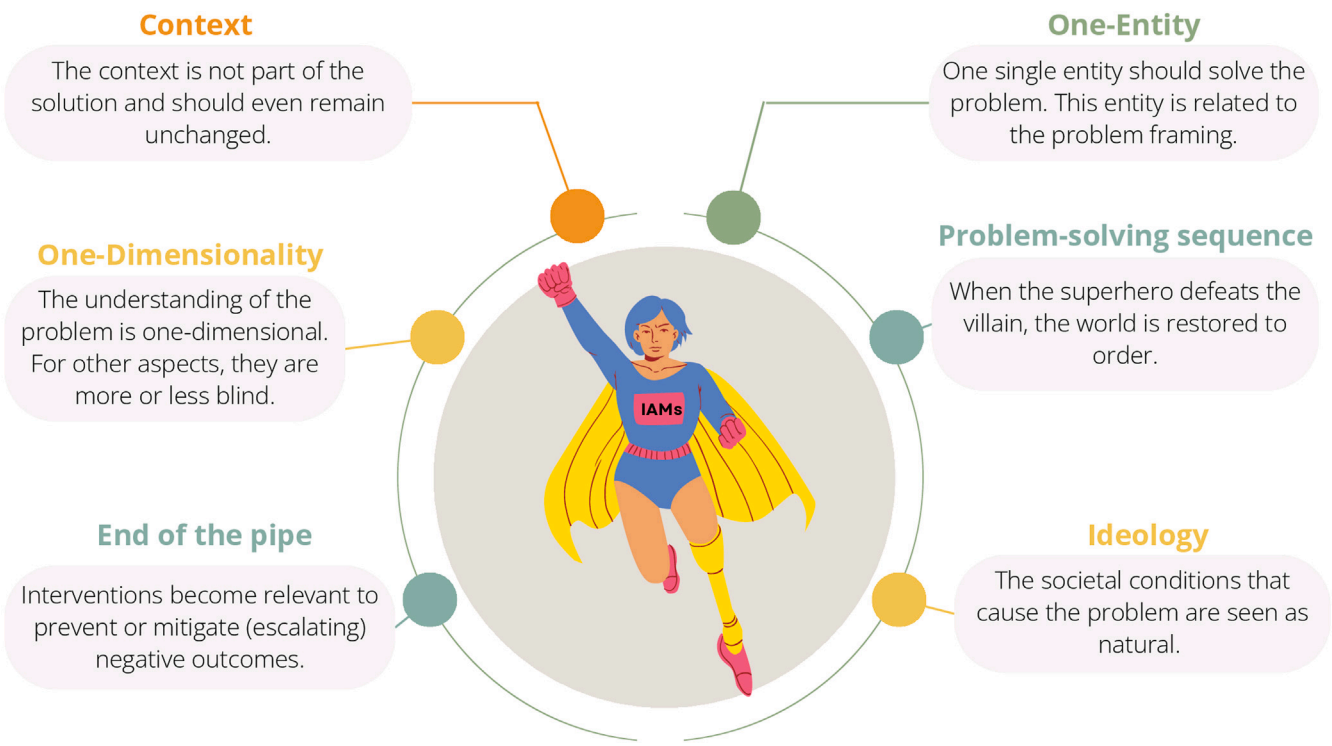

Fig. 1. Illustration of the superhero aspects represented in IAM research.

there is no alternative to the existing social conditions and that, with a few exceptions (the villains), they are basically good. IAM research does not explicitly include this social context in the scientific process and therefore activates the stabilizing functions. In turn, these then legitimize the existing societal conditions. This leads us to the practical crisis of IAM research. IAMs fail to achieve their goal of showing ways out of the climate crisis and instead contribute to the stabilization of crisis-ridden societal conditions.

Consequently, the superhero approach is not sufficient, as many interacting crises must be solved together. Therefore, all options for action must be seen as part of a Deep Transformation, which aims for long-term solutions that target changes in all societal areas. Hence, we understand a Deep Transformation as the reorganization of all aspects of life, including culture, human-nature relationships, economy, ethics, gender relations, and more. This far-reaching assumption is strongly justified, given that almost all modeled mitigation pathways rely on enormous amounts of future negative emissions. Whether negative emissions on such a scale are feasible and meaningful is highly debated [13,17–21]. Moreover, not only is the temperature limit of the Paris Agreement about to be exceeded [61] but also other planetary boundaries [62] and social boundaries [63].

4.3. The role of ideology

Based on the reviewed literature we demonstrate the ideological and normative influence on the economic foundations of IAMs (Section 4.3.2) and relate them to epistemic violence (Section 4.3.1). Through the critique of ideology, the realm of possibilities can be expanded again because it liquefies the existing, or in other words, shows that it is changeable. However, the indication of changeability does not provide criteria for why certain things should be changed [7]. Critique of ideology is therefore complete only when it indicates how to differentiate appropriate from problematic determinations of the possibility space [7, p. 282]. We address this requirement in 4.4 where we discuss two ways the realm of possibilities can be appropriately determined (Categorical Utopia and the Manoa Mash-up Method).

4.3.1. Epistemic violence

In the aforementioned co-production process, new political and social norms are simultaneously produced and solidified. If this is not sufficiently taken into account, the produced knowledge can reproduce injustices and constitute a form of violence. This is referred to as epistemic violence [77] understood as using "knowledge" to either oppress the legitimacy of the voices uttered by others, who do not have the same entitlement to claim that their beliefs and convictions qualify as (scientific) knowledge or devalue and sanction particular aspects of knowledge as less justified or morally sound.

Models are always based on a certain level of abstraction, which can lead to the loss of context. Regarding global models, Hulme [78] emphasizes the emergence of globalized knowledge. This knowledge has no connection to the location-specific realities of those affected. The focus on such global variables constructs the problem of the climate crisis more as an issue of rising temperature or CO_2 concentration. This understanding is also reflected in the Paris Agreement or the Copenhagen Accord. In contrast, a look at the Cochabamba Peoples Agreement shows how global environmental issues are described in a different way [78]. Knowledge is contextualized, meaning it revolves around, for example, people's rights to seeds, land, water, and food "through forms of production that are in harmony with Mother Earth and appropriate to local cultural contexts" [79].

The epistemic violence of Eurocentric knowledge has been and remains an essential part of colonialism [80,81]. For instance, Bluwstein and Cavanagh [82] illustrate the complex land conflicts that could arise in the Global South within the framework of scenarios compatible with the 1.5-degree target. Nevertheless, epistemic violence goes beyond globalized knowledge and the power relations of colonialism. This is because numerous assumptions about humans, society, and the problem are made in the models [9]. The foundation is laid in the first step of model formation by questioning what the problem is and what is important [35].

4.3.2. Ideological foundations

With few exceptions (e.g., Degrowth scenarios), IAMs are based on the assumptions of neoclassical economics. This economic framework

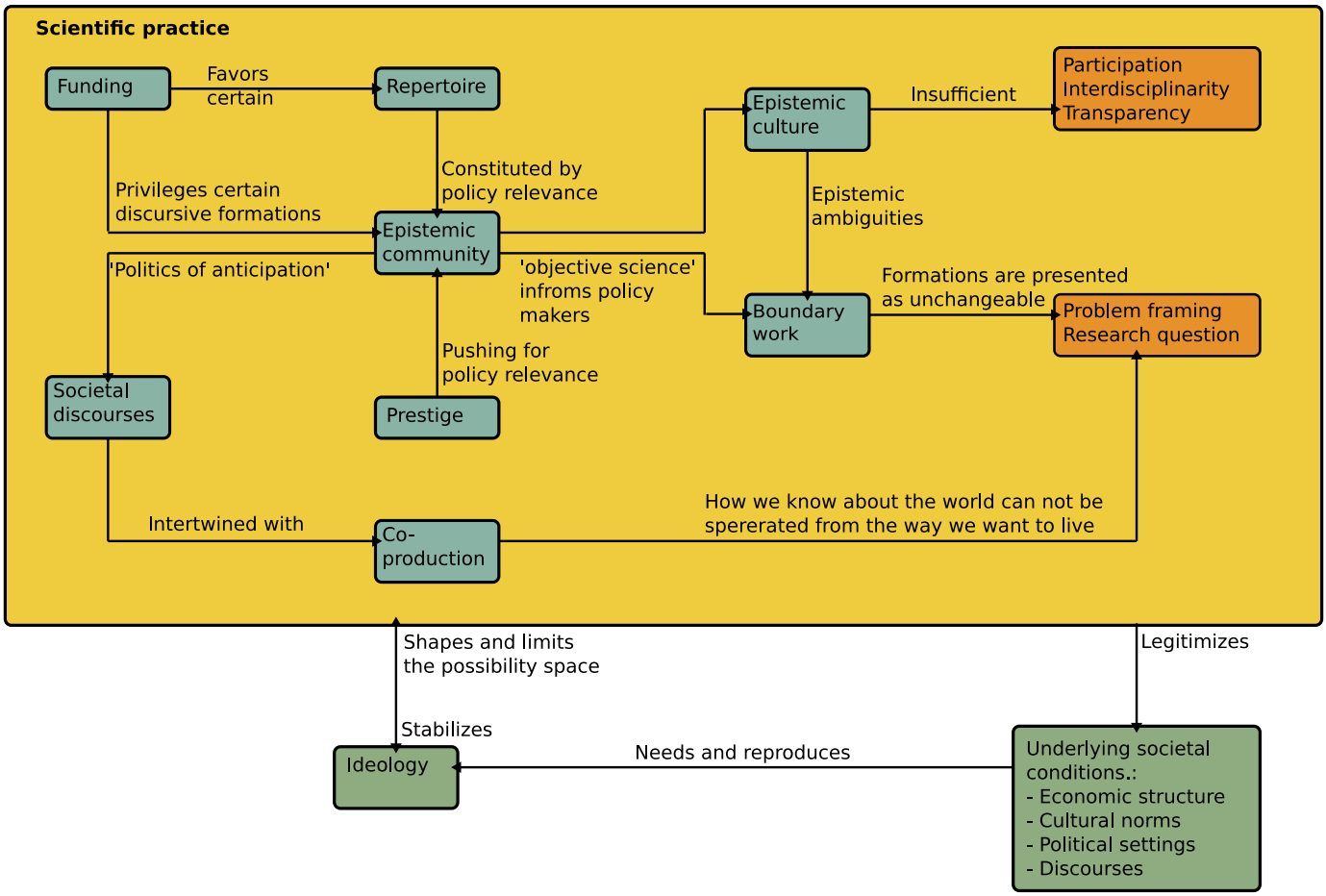

Fig. 2. Schematic representation of the interacting concepts that describe the scientific practice of modeling and their interaction with ideology and societal formations.

relies on a liberal egalitarian, market-based, or utilitarian understanding of justice [13]. In other words, questions of justice are usually explained or justified through individual performance or market mechanisms [13]. It is, therefore, logical that policy recommendations are tied to economic growth, fully functioning and perfect markets, and competitive thinking [13]. Within neoclassical economics, various and highly controversial assumptions must be made. An overview of these assumptions can be found in Table 1. Due to the value-based judgments and preferences necessary when choosing economic parameters, these are referred to as ethical parameters [35,83]. Beck and Krueger [35] consider these as intrinsic to models since they are often "hardwired" (p. 632) into them. Beck and Krueger [35] advocate for making such assumptions transparent and then subjecting them to the discussion.

In addition to these assumptions, criticism is also directed at the overall picture or the underlying perceptual model on which IAMs rely. "The perceptual model underlying IAMs is the structured and qualitative understanding of the climate change problem, including its causes, processes, and consequences" ([35], p. 636). The application of IAMs always requires problem framing, which is inherently ethical as it involves value judgments about what is considered relevant and what is not [35,84]. For instance, the cost–benefit approach determines when and how much mitigation is reasonable, instead of establishing this under ethical, political, and scientific judgments and then finding a cost-effective method [85]. Furthermore, problem framing already includes approaches to how a problem could be solved. IAMs typically adopt the market failure framing, attributing the climate crisis to a market failure where emissions are not (properly) priced [35]. According to Storm [84], this is the hegemonic discourse regarding IAMs. Thus, it is possible to incorporate emissions as externalities into economic standard models [35]. This framing cements carbon pricing as a solution and simultaneously assumes that all emissions are of equal value [35]. However, carbon pricing is not a neutral mechanism and is sharply criticized by Indigenous communities and social movements (e.g. [79]). Additionally, market competition always leads to the externalization of costs. This is not the exception but, according to Harvey and Luxemburg, a necessary aspect of a capitalist economy, functioning in the form of "accumulation through dispossession" [82,86,87]. Therefore, Stoddard et al. [47] argue that "climate change is not the result of a market failure but rather the outcome of a fully functioning capital accumulating economy working hard to shift costs onto others (85), and especially those who lack voice or power (such as the poor, future generations, children, and nonhumans)" (p. 666).

Describing the problem through such a reductionist lens in the form of carbon pricing, cost–benefit, and the Homo Economicus as a rational and self-interested human within national governance regimes reproduces this logic. At the same time, cooperative, solidarity-based, and justice-oriented solutions that transcend national borders are rendered irrational. Consequently, certain mitigation options are marked as rational, while others are labeled as irrational [19]. This places certain worldviews and ways of knowing above others.

4.4. Making IAMs ready

4.4.1. Utopia first!

In contrast to the superhero approach, the systematic examination of utopias enables a holistic view of society and developing alternative visions of the future that not only solve individual problems but also question the underlying societal conditions. The world-making power attributed to IAMs can consequently also make these futures accessible

Table 2
Stabilizing functions.

Concept	Stabilizing function
Epistemic community	Scientific communities can be identified by a common practice or around a knowledge structure. In both cases, participants have certain underlying assumptions about how a research question can be answered. These are historically and sociologically shaped. This epistemic culture also influences the modeling process [1,19,28,35,43].
Funding	Funding favors certain research (e.g., scientifically quantitative research; policymakers prefer research teams operating within the same discursive formation as they do) [13,14,25,35].
Co-production	Knowledge does not arise in a vacuum and is embedded in social practices, identities, norms, conventions, discourses, instruments, and institutions and, in turn, embeds them [9,22–25].
Boundary work	Boundary work refers to the processes of drawing boundaries between domains like "science" and "politics" or "society" and "nature." These boundaries legitimize certain knowledge as fixed or natural while excluding others [24,32,64].
World-making power	IAMs, pathways, and scenarios are not just tools but also have agency, mobilizing powerful future scenarios that impact reality [23,24,35].
Politics of anticipation	When scientists are asked to examine policy options, it has societal influence, as science is often viewed as a neutral entity with great authority and performativity [9,24].
Epistemic ambiguities	IAMs represent the present and the future while providing action options for policymakers. The relationship between the model and reality is understood differently. The influence of this community allows it to provide "robust" numbers for other disciplines [42,43,65,66].
Politics of prevarication	(No. yet) existing technologies to reduce atmospheric CO_2 concentration are perceived as a sensible option through examination with IAMs. This allows policymakers an escape, creating the impression that more effective policy options will be available in the future [23].
Insufficient transparency	IAMs are not transparent regarding normative assumptions, values, and uncertainties. Therefore, the results of different IAMs are difficult to compare [29,30,39,67,68].
Insufficient interdisciplinarity	Interdisciplinarity rarely extends to the social, humanities, or political sciences, although this collaboration would be crucial for investigating a Deep Transformation [10,19,31,34,45,69–72].
Insufficient participation	For meaningful assessment, stakeholders should ideally be involved early in the research process. This way, values or local knowledge can be incorporated [13,68,69,73–76].

and thus politically negotiable.

A transformation always has a target state, even if this is often not made explicit. It is only through a debate about possible utopias and their scientific feasibility that it becomes apparent that there are various competing utopias. Therefore, research on transformations including IAM modeling has to first address the question of the goal (utopia question) and then the question of how to achieve it (transformation question) [5]. By prioritizing utopia, the previous use of the future as a "colonial outpost" ([18], p. 8) where negative externalities can be shifted, akin to a colony, is counteracted by design.

We can envision a different world, but only from today's perspective. However, if conditions change fundamentally, people change as well, and alternative futures become conceivable [6]. Therefore, researching alternative futures can only be a process towards Deep Transformations. However, this process can gradually make other futures conceivable. In this way, emancipatory movements can be supported when they are accused of having an impossible vision of the future.

4.4.2. Whose future? — From listening to the science to listening to the movement(s)

IAMs project the existing societal conditions into the future, thus limiting the space for radical social change through their stabilizing functions. In this sense, some argue that research models should be used to explore alternative futures rather than narrowing down the decision space [9,10,16]. Even though post-growth scenarios have been examined neglectfully by IAM research, they illustrate how IAMs, can explore alternative futures. On the other hand, neglecting alternative futures perpetuates a self-fulfilling prophecy, hindering, for example, the legitimacy of degrowth scenarios [88].

However, these scenarios are based on assumptions that scientists consider possible or conceivable and differ for instance from land-rights-based mitigation, as demanded by Indigenous communities [79,89]. Other authors, such as van Beek et al. [10], explicitly argue for expanding scenarios using grassroots and bottom-up approaches. The phenomenon of Co-production can be used in an emancipatory manner to counteract epistemic violence by explicitly involving non-scientists in knowledge creation. This requires scientists to be aware not only of the epistemic but also of the political, social, and cultural values intertwined with their research [18,22]. Such an active debate is already being implemented, for example, in critical physical geography [90,91]. This discourse seeks not only interdisciplinary integration but also the cultivation of a critical practice, wherein scientific assertions must be defended with reference to values [92].

The incorporation of different worldviews can be considered in a participatory process iteratively (e.g., through participatory world-building [73]) [10,16,71]. A promising example of participatory scenario development is the Manoa Mash-Up Method [74] based on the Seeds approach [93]. In this approach, a seed represents something that plays a small role today but could develop and play a significant role in the future. The impacts of the fully grown seeds and their interactions are then examined and woven into a story. This leads to the development of pathways to reach this future, detailing what is necessary and what needs to change. Through this method, a utopia can be developed that does not arise from fantasy but from the maturation of approaches (seeds) that already exist today. Through such a participatory approach, a paradigm shift from the slogan of the Fridays for Future movement "Listen to the Science" to "Listen to the Movements" is possible. Thus, scientific studies can consider the significantly more radical future visions of social movements. In this way, the modeling space would not be limited by the dominant standards or those of individual modelers but by the specific and visible social actors.

However, participation alone is not the sole solution, as it is still constrained by institutional conditions, largely based on cultures and institutions of the Global North [1,13]. Collaboration should not only be with the social sciences [13] but also with the arts to approach creative possible futures [16]. These futures should not only be technically represented in the form of plots and graphics but made experiential through all the senses [16]. In any case, participatory model development is based on conceptions of humanity and society through the envisioned future visions derived from them. Even in a participatory approach, such as the Manoa Mash-Up Method, it is essential to make these visible and check them for theoretical consistency. We illustrate how this can be systematically done in the next section.

4.4.3. Categorical Utopia

We understand the goal of a Deep Transformation as overcoming the crisis-ridden societal conditions. The dynamics of those multiple crises can be explained in terms of certain scientifically derived categories [5]. Based on categorical criticism, a Categorical Utopia can be designed that breaks with existing categories. Whether the Categorical Utopia is possible can be scientifically investigated. The basis for this is the exposure of the theory about the individual and society on which the Categorical Utopia is based [5]. The Categorical Utopia can then be criticized on three levels [5]:

- Critique of the underlying social and individual theory
- Critique of the relationship between this theoretical basis and the utopia executed in content (e.g., Is the utopia covered by the theory?)
- Critique of the conceptual development within the Categorical Utopia (e.g., are implicit assumptions made?)

The Categorical Utopia framework is primarily concerned with political and social feasibility, neglecting the natural scientific aspects [5]. Here lies the opportunity for us to explore Deep Transformations through IAMs by examining the interaction of a Categorical Utopia with non-human nature.

5. Conclusion

Many people enjoy the thrill of superhero movies, but when it comes to tackling real interacting crises, we need a more grounded approach than what we see on the silver screen. With this in mind, we propose taking the initial steps towards a comprehensive exploration of Deep Transformations using IAMs. Our first objective is to assess the capabilities of existing IAMs and their sub-models in delving into Deep Transformations within the framework of the Categorical Utopia. Secondly, models need to be adapted accordingly, using a participatory approach such as the Manoa Mash-up Method. This can be tested with sub-models such as land use models, while the scientific practice can draw significant inspiration from the pioneering approaches of critical physical geography, particularly in terms of fostering stakeholder engagement and embracing critical interdisciplinarity. However, in the long term, it might be useful to engage with frameworks that are less concrete and go beyond the Categorical Utopia framework such as Utopia as a Method [94].

IAM research has garnered significant attention from the social sciences due to its crucial policy implications, resulting in a rich body of critical literature. Nevertheless, we firmly believe that our insights extend beyond the realm of IAM research, offering valuable contributions to other model families as well.

Finally, the superhero metaphor was employed as a widely recognized and accessible framing device to convey our findings, rather than as a fully developed conceptual framework. Future research could explore which superhero definition from literature studies best aligns with IAMs.

CRediT authorship contribution statement

Felix Krawczyk: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Andreas Ch. Braun:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Acknowledgments

We thank Rebecca Lave for stimulating discussions and feedback.

Funding

This research was supported by a Ph.D. scholarship from the Hans-Böckler-Foundation awarded to F.K.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) in order to improve the readability and language of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Felix Krawczyk reports financial support was provided by Hans-Bockler-Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary information is available in the online version of the paper. It includes a detailed description of the method as well as the PRISMA Flow Diagram.

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.erss.2025.103959.

Data availability

Data will be made available on request.

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